Kejun "Albert" Ying, Ph.D.

290 Jane Stanford Way, Stanford, CA 94305
keying@stanford.edu | kying0@uw.edu albert-ying 0000-0002-1791-6176 kejunying.com

Studying aging at the intersection of biology and AI

Education & Experience

Stanford University & University of Washington — Stanford, CA & Seattle, WA
Postdoctoral Researcher — Spring 2025 – Present
- Co-advised by Dr. Tony Wyss-Coray and Dr. David Baker
- NIH/NIA F99/Koo Fellow

Harvard University — Cambridge, MA
Ph.D., Biological Science in Public Health — Fall 2019 – Spring 2025
- Dissertation: "On the Quantification of Aging"
- Advisor: Dr. Vadim Gladyshev, Harvard Medical School, Brigham and Women's Hospital

Harvard University — Cambridge, MA
M.S., Computational Science Engineering — Summer 2023 – Spring 2024
- Secondary field during Ph.D. study

University of California, Berkeley — Berkeley, CA
Visiting Student, Integrative Biology — Summer 2017 – Fall 2017

Sun Yat-Sen University (Yat-Sen Honor School) — Guangzhou, China
B.S., Life Science — Fall 2015 – Summer 2019
- Thesis: Screening for the Interactome of hTERC based on Molecular Fluorescence Complementation System in Living Cells
- Advisor: Dr. Zhou Songyang

Other Professional Experience

Stanford University & University of Washington — Stanford, CA & Seattle, WA
Visiting Scholar, Wyss-Coray's Lab & Baker's Lab — Nov 2024 – May 2025

Harvard Medical School, Boston Children's Hospital — Boston, MA
RNA Modifications *Graduate Researcher (Rotation), Eric Greer's Lab* — Jan 2020 – Mar 2020

Harvard Medical School — Boston, MA
Cell Reprogramming *Graduate Researcher (Rotation), David Sinclair's Lab* — Oct 2019 – Dec 2019

Harvard T. H. Chan School of Public Health — Boston, MA
mTORC1 *Graduate Researcher (Rotation), Brendan Manning's Lab* — July 2019 – Oct 2019

Undergraduate Research — 2015 – 2019

- **Sun Yat-Sen University**, *Telomere & Telomerase* — Zhou Songyang's Lab
- **University of Edinburgh**, *Population genetics* — Xia Shen's Lab
- **University of Washington**, *Acarbose & Raparycin* — Matt Kaeberlein's Lab
- **Buck Institute for Research on Aging**, *Senolytics* — Judith Campisi's Lab
- **University of California, Berkeley**, *SIRT7* — Danica Chen's Lab

- **Sun Yat-Sen University**, *Telomere & DNA Methylation* Yikang Rong's Lab

Grants

THRIVE: Transforming Health: Reclaiming Intrinsic Vitality for Everyone ARPA-H PROSPR

Developing PROSPR Intrinsic Capacity score for predicting 20-year health outcomes, 2026 – 2031

- USD $36 million over 5 years (Key Personnel)
- Multi-institutional collaboration: Stanford University, Harvard University, Buck Institute for Research on Aging
- Integrating multi-omics biomarkers and functional assessments across 19+ longitudinal databases to enable personalized health monitoring and accelerate clinical trials

Aging and Longevity Award ResearchHub Foundation RFP

Funded proposal selected through open peer review, 2026 – 2027

OpenAI Researcher Access Program OpenAI

Research credit grant supporting AI-driven aging biology and protein design, 2026

- USD $20,000 in API credits

Koo Post-doctoral Transition Award NIH/NIA

Using causal aging biomarkers and protein design to develop novel anti-aging interventions, 2025 – 2028

F99 Transition to Aging Research for Predoctoral Students NIH/NIA *Using causal aging biomarkers and protein design to develop novel anti-aging interventions*, 2024 – 2025

- Award Document Number: FAG088431A (PI)
- Received a *perfect* Impact Score of **10**

Publications

† Corresponding author; * Co-first author; + Contributed as consortium author

Selected Publications & Preprints

Wang, Z., Zhang, O., Wang, J., Jiang, Y., Ma, Z. C., ..., Gladyshev, V. N., Wyss-Coray, T., & **Ying, K.**† (2026). A world model simulates the latent dynamics of human health. **medRxiv**. https://doi.org/10.64898/2026.09.19.26363460

Ying, K., Paulson, S., Reinhard, J., ..., Gladyshev, V. N. (2026). An Open Competition for Biomarkers of Aging. **Nature Aging**, *6*, 1193–1195. https://doi.org/10.1038/s43587-026-01139-6

Moqri, M.*, **Ying, K.***, Poganik, J.*, Herzog, C.*, ..., Marioni, R. E., Lasky-Su, J. A., Snyder, M. P., & Gladyshev, V. N. (2026). Integrative epigenetics and transcriptomics identify aging genes in human blood. **Nature Communications**, *16*, 67369. https://doi.org/10.1038/s41467-025-67369-1

Ying, K., Paulson, S., Eames, A., Tyshkovskiy, A., ..., Gladyshev, V. N. (2025). *A Unified Framework for Systematic Curation and Evaluation of Aging Biomarkers*. **Nature Aging**. https://www.nature.com/articles/s43587-025-00987-y

Wu, X.*, Liu, H.*, **Ying, K.***† (2025). Biological Age, Aging Clocks, and the Interplay with Lymphoid Neoplasms: Mechanisms and Clinical Frontiers. **Lymphatics**, *3*(3), 19. https://doi.org/10.3390/lymphatics3030019

Ying, K.† (2024). Causal inference for epigenetic ageing. **Nature Reviews Genetics**, 1–1. https://doi.org/10.1038/s41576-024-00799-7

Ying, K., Castro, J. P., Shindyapina, A. V., ..., Gladyshev, V. N. (2024). Depletion of loss-of-function germline mutations in centenarians reveals longevity genes. **Nature Communications**, *15*(1), 5956. https://doi.org/10.1038/s41467-024-52967-2

Ying, K., Liu, H., Tarkhov, A. E., ..., Gladyshev, V. N. (2024). Causality-enriched epigenetic age uncouples damage and adaptation. **Nature Aging (February Cover)**, 1–16. https://doi.org/10.1038/s43587-023-00557-0

Ying, K., Zhai, R., Pyrkov, T. V., ..., Gladyshev, V. N. (2021). Genetic and phenotypic analysis of the causal relationship between aging and COVID-19. **Communications Medicine**, *1*(1), 35. https://doi.org/10.1038/s43856-021-00033-z

Ying, K.,*† Tyshkovskiy, A., Moldakozhayev, A., ..., Gladyshev, V. N. (2025). Autonomous AI Agents Discover Aging Interventions from Millions of Molecular Profiles. **bioRxiv (Nature under review)**. https://doi.org/10.1101/2023.02.28.530532

Ying, K.†**,** Song, J., Cui, H., ..., Gladyshev, V. N†. (2024). MethylGPT: a foundation model for the DNA methylome. **bioRxiv (Nature Aging 1st Revision)**. https://doi.org/10.1101/2024.10.30.621013

Ying, K., Tyshkovskiy, A., Chen, Q., ..., Gladyshev, V. N. (2024). High-dimensional Ageome Representations of Biological Aging across Functional Modules. **bioRxiv (Nature Aging 2nd Revision)**. https://doi.org/10.1101/2024.09.21.570935

Other Publications

Zhavoronkov, A., Galkin, F., Chen, S., ..., **Ying, K.**, ..., Gladyshev, V. N. (2026). Integration of proteomic aging clocks in a phase 2a clinical trial supports simultaneous geroprotective assessment. **Nature Biotechnology**. https://doi.org/10.1038/s41587-026-03286-y

Kordowitzki, P., & **Ying, K.** (2026). Novel Exploratory Transcriptomic Candidates as Biomarkers and Cancer Hallmark Fingerprints for Ovarian Endometroid and Clear Cell Carcinomas in Women. **Antioxidants**, *15*(8), 979. https://doi.org/10.3390/antiox15080979

Wu, F., Xuan, W., Qi, H., ..., **Ying, K.,** ..., Choi, Y. (2026). Proteo-R1: Reasoning Foundation Models for De Novo Protein Design. **ICML 2026**. https://arxiv.org/abs/2605.02937

Zhavoronkov, A., **Ying, K.,** & Wilczok, D. (2026). Peakspan: Defining, Quantifying and Extending the Boundaries of Peak Productive Lifespan. **Aging and Disease**. https://doi.org/10.14336/AD.2026.0080

Kordowitzki, P., & **Ying, K.** (2026). The pursuit of understanding human longevity. **npj Aging**, *12*(1), 25. https://doi.org/10.1038/s41514-026-00339-z

Tyshkovskiy, A., Kholdina, D., **Ying, K.,** Davitadze, M., ..., Gladyshev, V. N. (2026). Universal transcriptomic hallmarks of mammalian ageing and mortality. **Nature**, *654*, 173–188. https://doi.org/10.1038/s41586-026-10542-3

Mavrommatis, C., Belsky, D. W., **Ying, K.,** Moqri, M., Campbell, A., Richmond, A., Gladyshev, V. N., Chandra, T., McCartney, D. L., & Marioni, R. E. (2025). An unbiased comparison of 14 epigenetic clocks in relation to 174 incident disease outcomes. **Nature Communications**, *16*, 11164. https://doi.org/10.1038/s41467-025-66106-y

Zhang, O., Lin, H., Zhang, X., Wang, X., Wu, Z., Ye, Q., Zhao, W., Wang, J., **Ying, K.,**, Kang, Y., Hsieh, C.-Y., Hou, T. (2025). Graph neural networks in modern AI-aided drug discovery. **Chemical Reviews**, 125, 10001–10103. https://doi.org/10.1021/acs.chemrev.5b00254

Zhang, O., ..., **Ying, K.,** Huang, Y., Zhao, H., Kang, Y., Pan, P., Wang, J., Guo, D., Zheng, S., Hsieh, C.-Y., & Hou, T. (2025). ECloudGen: leveraging electron clouds as a latent variable to scale up structure-based molecular design. **Nature Computational Science**. https://doi.org/10.1038/s43588-025-00886-7

Farinas, A., Rutledge, J., Bot, V. A., Western, D., **Ying, K.,** Lawrence, K. A., Oh, H. S. H., ..., Wyss-Coray, T. (2025). Disruption of the cerebrospinal fluid–plasma protein balance in cognitive impairment and aging. **Nature Medicine**, 1–12. https://doi.org/10.1038/s41591-025-03831-3

Rothi, M.H., Sarkar, G.C., Haddad, J.A., Mitchell, W., **Ying, K.,** et al. (2025). The 18S rRNA methyltransferase DIMT-1 regulates lifespan in the germline later in life. **Nature Communications**, 16, 6944. https://doi.org/10.1038/s41467-025-62323-7

Grzeczka, A., Iqbal, S., **Ying, K.,** Kordowitzki, P. (2025). Circular RNAs as regulators and biomarkers of mammalian ovarian ageing. **GeroScience**, 1–19. https://doi.org/10.1007/s11357-025-01798-0

Jacques, E., Herzog, C., **Ying, K.,** ... Gladyshev, V. N. (2025). Invigorating discovery and clinical translation of aging biomarkers. **Nature Aging**, 1–5. https://doi.org/10.1038/s43587-025-00838-w

Goeminne, L. J. E., Vladimirova, A., Eames, A., Tyshkovskiy, A., Argentieri, M. A., **Ying, K.,** Moqri, M., & Gladyshev, V. N. (2025). Plasma protein-based organ-specific aging and mortality models unveil diseases as accelerated aging of organismal systems. **Cell Metabolism**, https://doi.org/10.1016/j.cmet.2024.10.005

Gladyshev, V. N., Anderson, B., Barlit, H., ..., **Ying, K.,** Yunes, J., Zhang, B., & Zhavoronkov, A. (2024). Disagreement on foundational principles of biological aging. **PNAS Nexus**, *3*(12), pgae499. https://doi.org/10.1093/pnasnexus/pgae499

Lyu, YX.*, Fu, Q.*, Wilczok, D.*, **Ying, K.***, King, A., ..., Bakula, D. (2024). Longevity biotechnology: Bridging AI, biomarkers, geroscience and clinical applications for healthy longevity. **Aging**, *16*(1), 1–25. https://doi.org/10.18632/aging.206135

Biomarkers of Aging Consortium+, Herzog, C. M. S., Goeminne, L. J. E., Poganik, J. R., ..., Gladyshev, V. N. (2024). Challenges and recommendations for the translation of biomarkers of aging. **Nature Aging**, 1–12. https://doi.org/10.1038/s43587-024-00683-3

Castro, J. P., Shindyapina, A. V., Barbieri, A., **Ying, K.,** ..., Gladyshev, V. N. (2024). Age-associated clonal B cells drive B cell lymphoma in mice. **Nature Aging**, *4*(8), 1–15. https://doi.org/10.1038/s43587-024-00671-7

Moqri, M., ..., de Sena Brandine, G., **Ying, K.,** Tarkhov, A., ..., Sebastiano, V. (2024). PRC2-AgeIndex as a universal biomarker of aging and rejuvenation. **Nature Communications**, *15*(1), 5956. https://doi.org/10.1038/s41467-024-50098-2

Tarkhov, A. E., Lindstrom-Vautrin, T., Zhang, S., **Ying, K.,** Moqri, M., ..., Gladyshev, V. N. (2024). Nature of epigenetic aging from a single-cell perspective. **Nature Aging**, 1–17. https://doi.org/10.1038/s43587-023-00555-2

Moqri, M., Herzog, C., Poganik, J. R., **Ying, K.,**, ... Ferrucci, L. (2024). Validation of biomarkers of aging. **Nature Medicine**, 1–13. https://doi.org/10.1038/s41591-023-02784-9

Griffin, P. T., ..., Kerepesi, C., **Ying, K.,** ..., Sinclair, D. A. (2024). TIME-seq reduces time and cost of DNA methylation measurement for epigenetic clock construction. **Nature Aging**, 1–14. https://doi.org/10.1038/s43587-023-00555-2

Moqri, M., Herzog, C., Poganik, J. R., **Biomarkers of Aging Consortium**+, ... Gladyshev, V. N. (2023). Biomarkers of aging for the identification and evaluation of longevity interventions. **Cell**, *186*(18), 3758–3775. https://doi.org/10.1016/j.cell.2023.08.003

Liberman, N., Rothi, M. H., Gerashchenko, M. V., Zorbas, C., Boulias, K., MacWhinnie, F. G., **Ying, A. K.,** Flood Taylor, A., ..., Greer, E. L. (2023). 18S rRNA methyltransferases DIMT1 and BUD23 drive intergenerational hormesis. **Molecular Cell**, *83*(18), 3268–3282.e7. https://doi.org/10.1016/j.molcel.2023.08.014

Bitto, A., Grillo, A. S., Ito, T. K., Stanaway, I. B., Nguyen, B. M. G., **Ying, K.,** ... Kaeberlein, M. (2023). Acarbose suppresses symptoms of mitochondrial disease in a mouse model of Leigh syndrome. **Nature Metabolism**, *5*(6), 955–967. https://doi.org/10.1038/s42255-023-00815-w

Emmrich, S., Trapp, A., Tolibzoda Zakusilo, F., Straight, M. E., **Ying, A. K.,** Tyshkovskiy, A., ..., Gorbunova, V. (2022). Characterization of naked mole-rat hematopoiesis reveals unique stem and progenitor cell patterns and neotenic traits. **The EMBO Journal**, *41*(15), e109694. https://doi.org/10.15252/embj.2021109694

Yang, Z., ..., Guo, H., **Ying, K.,** Gustafsson, S., ..., Shen, X. (2022). Genetic Landscape of the ACE2 Coronavirus Receptor. **Circulation**, *145*(18), 1398–1411. https://doi.org/10.1161/CIRCULATIONAHA.121.057888

Li, T., Ning, Z., Yang, Z., Zhai, R., Zheng, C., Xu, W., Wang, Y., **Ying, K.,** Chen, Y., & Shen, X. (2021). Total genetic contribution assessment across the human genome. **Nature Communications**, *12*(1), 2845. https://doi.org/10.1038/s41467-021-23124-w

Zhu, J., Xu, M., Liu, Y., Zhuang, L., **Ying, K.,** Liu, F., ..., Songyang, Z. (2019). Phosphorylation of PLIN3 by AMPK promotes dispersion of lipid droplets during starvation. **Protein & Cell**, *10*(5), 382–387. https://doi.org/10.1007/s13238-018-0593-9

Other Preprints

Zhang, S., Tyshkovskiy, A., **Ying, K.,** Wang, S., & Gladyshev, V. N. (2026). Mammalian aging involves genome-wide splicing degeneration leading to functional decline. **bioRxiv**. https://doi.org/10.64898/2026.06.26.734787

Eames, A., Glubokov, D., Moldakozhayev, A., Yücel, A. D., Tyshkovskiy, A., **Ying, K.,** Goeminne, L. J. E., de Magalhães, C. G., & Gladyshev, V. N. (2026). Anatomy of aging through organ-resolved multi-modal imaging and deep learning. **medRxiv**. https://doi.org/10.64898/2026.03.14.26348392

Roberts, K. F., Abrams, Z. B., Cappelletti, L., Moqri, M., ..., **Ying, K.,** ..., Reese, J. T. (2026). OpenScientist: evaluating an open agentic AI co-scientist to accelerate biomedical discovery. **medRxiv**. https://doi.org/10.64898/2026.03.15.26348338

Wu, Y. C., Yin, M., Shi, B., Zhang, Z., ..., **Ying, K. A.,** ..., Wang, M., & Cong, L. (2026). MedOS: AI-XR-Cobot World Model for Clinical Perception and Action. **medRxiv**. https://doi.org/10.64898/2026.02.18.26345936

Zhang, O., Zhang, X., Lin, H., Tan, C., Wang, Q., Mo, Y., ..., **Ying, K.,** Li, J., Zeng, Y., Lang, L., Pan, P., Cao, H., Song, Z., Qiang, B., Wang, J., Ji, P., Bai, L., Zhang, J., Hsieh, C.-Y., Heng, P. A., Sun, S., Hou, T., & Zheng, S. (2025). ODesign: A World Model for Biomolecular Interaction Design. **arXiv**. https://odesign1.github.io/

Mavrommatis, C., Belsky, D., **Ying, K.,** Moqri, M., Campbell, A., Richmond, A., ..., Gladyshev, V. N. (2025). *An unbiased comparison of 14 epigenetic clocks in relation to 10-year onset of 174 disease outcomes in 18,859 individuals*. **medRxiv**. https://doi.org/10.1101/2025.07.14.25331494

Galkin, F., ..., Tyshkovskiy, A., **Ying, K.,** Gladyshev, V. N., & Zhavoronkov, A. (2024). Precious3GPT: Multimodal Multi-Species Multi-Omics Multi-Tissue Transformer for Aging Research and Drug Discovery. **bioRxiv**. https://doi.org/10.1101/2024.07.25.605062

Rothi, M. H., Sarkar, G. C., Al Haddad, J., Mitchell, W., **Ying, K.,** Pohl, N., ..., Greer, E. L. (2024). The 18S rRNA Methyltransferase DIMT-1 Regulates Lifespan in the Germline Later in Life. **bioRxiv**. https://doi.org/10.1101/2024.05.15.570935

Zhang, B., Tarkhov, A. E., Ratzan, W., **Ying, K.,** Moqri, M., ..., Gladyshev, V. N. (2022). *Epigenetic profiling and incidence of disrupted development point to gastrulation as aging ground zero in Xenopus laevis*. **bioRxiv**. https://doi.org/10.1101/2022.08.02.502559

Patents

V. N. Gladyshev, **K. Ying**, "High-dimensional measurement of biologicar age" (2024). *Provisional Patent Application*

V. N. Gladyshev, **K. Ying**, "Mapping CpG sites to quantify aging traits" (2024). *WO2024039905A2*

Software and Database

Autonomous Lab (2026) https://github.com/albert-ying/autonomous-lab
MCP server for multi-agent autonomous workflows; AI agents execute senior-junior loops while humans steer decisions.

ClockBase Agent (2025) https://www.clockbase.org/
AI agent system that autonomously discovers aging-modifying interventions from ~2M human and mouse molecular profiles.

MethylGPT (2024) https://github.com/albert-ying/MethylGPT
Foundation model for the DNA methylome, pretrained on 226K+ methylation profiles from 5,281 datasets.

Biolearn (2024) https://bio-learn.github.io/
Open-source Python library for biomarkers of aging analysis with reference implementations of common aging clocks.

ClockBase (2023) http://gladyshevlab.org:3838/ClockBase/
Comprehensive platform for biological age profiling across multiple clocks in human and mouse (~300K samples).

Presentations

Oral Presentations

Biomarkers of Aging Symposium 2025 Boston, MA
Massive AI agent mining of aging-modifying interventions from millions of molecular profile 2025

ASHG 2025 Annual Meeting Boston, MA
Decoding the Aging Methylome: From Causal Inference to Foundation Models 2025

6th TimePie Longevity Forum Shanghai, China
Massive AI agent mining of aging-modifying interventions from millions of molecular profile 2025

CSH-Asia Conference: Stem cell, Aging and Rejuvenation Suzhou, China
Massive AI agent mining of aging-modifying interventions from millions of molecular profile 2025

Keystone Symposia: Aging: New Frontiers in Rejuvenation and Gerotherapeutics Breckenridge, CO
MethylGPT: A Foundation Model for the DNA Methylome 2025

Biomarkers of Aging Symposium Boston, MA
Standardization of aging biomarkers and BoA challenge 2024

Harvard GRIP Presentations Boston, MA
Causal Aging Biomarker empowers Unbiased Anti-Aging Therapy Screening 2024

4th TimePie Longevity Forum Shanghai, China
Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening 2023

Global Congress on Aesthetic and Anti-Aging (GCAA2023) — Singapore
Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening — 2023

10th Aging Research and Drug Discovery conference (ARDD2023) — Copenhagen, Denmark
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

AGE 2023 51st Annual Meeting — Oklahoma City, OK
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Broad Institute MPG Retreat — Cambridge, MA
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Harvard GRIP Presentations — Boston, MA
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

Targeting Metabesity 2022, 'Honorable Mention' — Virtual Conference
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

GSA 2021 Annual Scientific Meeting — Virtual Conference
Genetic and phenotypic evidence for causal relationships between aging and COVID-19 — 2021

Invited Talks

St. Jude Children's Research Hospital, hosted by Dr. Zhaoming Wang — Memphis, TN
MethylGPT and Causality-enriched Epigenetic Clock — 2025

The Alliance for Longevity Initiatives Scientist Spotlight, — Online Podcast
Episode 14: Albert Ying — 2024

BioAge Seminar, hosted by Dr. Robert Hughes & Dr. Paul Timmers — Boston, MA
Ageome: Biological age with higher-dimensionality — 2024

MRC Integrative Epidemiology Unit Seminar — Bristol, UK
Epigenetic Clocks and Mendelian Randomization — 2024

NIA EL Projects Joint Meeting, National Institute on Aging — Online Webinar
Aging Clocks — 2024

Biomarkers of Aging Challenge, Foresight Institute — Online Webinar
Update Webinar with Foresight — 2024

Everything Epigenetics, podcast hosted by Hannah Went — Online Podcast
Causal Epigenetic Age Uncouples Damage and Adaptation — 2024

Chinese University of Hong Kong, hosted by Dr. Xin Wang — Hong Kong, China
Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening — 2024

Everything Epigenetics, podcast hosted by Hannah Went — Online Podcast
Causal Epigenetic Age Uncouples Damage and Adaptation — 2023

Chinese University of Hong Kong, hosted by Dr. Xin Wang — Hong Kong, China
Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening — 2023

Peking University, hosted by Dr. Jingdong Han — Beijing, China
Causal Aging Biomarker and ClockBase — 2023

Chinese Academy of Sciences, hosted by Dr. Xuming Zhou — Beijing, China
Causal Epigenetic Age Uncouples Damage and Adaptation — 2022

Foresight Institute, hosted by Allison Duettmann — Online Seminar
Genetic Variation, Aging & Relationship to COVID-19 | Joris Deelen, Albert Ying — 2020

Honors

WAIC Yunfan Award, "Tomorrow Star", World Artificial Intelligence Conference — 2026

Butler-Williams Scholar (Selected but unable to attend), National Institute on Aging (NIH) — 2026

Semifinalist, Harvard President's Innovation Challenge, Health Care and Life Sciences Track — 2025

Best Poster Award, Inaugural Biomarker of Aging Symposium — 2023

Best Poster Award, Gordon Research Conference, Systems Aging — 2022

Hackathon Winner, Longevity Hackathon, VitaDAO — 2021

Yat-Sen Honor School Program, Sun Yat-Sen University — 2016 – 2019

Yat-Sen Scholarship, Sun Yat-Sen University — 2016 – 2019

Service & Leadership

Board Member, Lifeboat Foundation — 2025 – Present

Member, Norn Longevity Nexus — 2025 – Present

Jury and Mentor, Agentic AI Against Aging Hackathon — 2025

Core Member, Biomarkers of Aging Consortium — 2024 – Present

Organizer, Biomarker of Aging Challenge — 2024 – Present

President, Harvard Interdisciplinary Discussion on Disease and Health — 2024 – 2025

Agenda Contributor, World Economic Forum — 2024

Organizing Committee Member, Biomarker of Aging Symposium 2024 — 2024

Organizing Committee Member, Biomarker of Aging Symposium 2023 — 2023

Teaching & Mentoring

Mentor, Yuanpei Young Scholars Program — 2023 – 2024

Instructor, Harvard Public Health Symposium For Young Generation — 2023

Students Supervised

Predoctoral Students: Ali Doga Yucel, Siyuan Li, Hanna Liu, Donghyun Lee, Yikun Zhang

Journals Reviewed

Nature, *Nature Aging*, *Nature Communications*, *Communications Medicine*, *Genome Medicine*, *BMC Nephrology*, *Scientific Reports*, ...

Selected Press & Media

Science — Eric Topol, "Learning the language of life with AI" Jan 2025
Featuring MethylGPT.

Nature Aging — "Epigenetic clock work ticks forward" Feb 2024
Expert commentary featuring causal epigenetic clocks.

Harvard Gazette — "Looking to rewind the aging clock" Feb 2024
Featuring causal epigenetic clocks.

World Economic Forum — "Not just a number: Science is redefining how we understand ageing" Nov 2024
Featuring aging clocks for global policy audience.

Nature Biotechnology — "Scientists hone tools to measure aging and rejuvenation interventions" Aug 2023
Featuring ClockBase platform for aging intervention screening.

Lifespan.io — "An AI-Based System Has Found a Potential Longevity Drug" Dec 2025
Featuring ClockBase Agent and ouabain discovery.

MIT Technology Review — "A 1990s-Born Chinese Scientist Sparks Anti-Aging Breakthrough" Dec 2025
Featuring AI agent-driven aging intervention discovery.

PÚBLICO (Portugal's leading newspaper) — "O que dizem os genes de quem tem 100 anos" Dec 2024 Featuring centenarian longevity genes.

Diário de Notícias (Portugal) — "Que segredos escondem os genes de quem vive até aos 100 anos?" Dec 2024

ScienceDaily — "New epigenetic clocks reinvent how we measure age" Feb 2024

References

Dr. Tony Wyss-Coray, Postdoctoral Co-Advisor twc@stanford.edu
D.H. Chen Distinguished Professor of Neurology and Neurological Sciences, Stanford University

Dr. David Baker, Postdoctoral Co-Advisor dabaker@uw.edu
Professor of Biochemistry, University of Washington

Dr. Vadim Gladyshev, Dissertation Advisor vgladyshev@bwh.harvard.edu
Professor of Medicine, Harvard Medical School

Dr. Steve Horvath, Collaborator shorvath@mednet.ucla.edu
Professor of Human Genetics, UCLA

Dr. David Sinclair, Dissertation Advisory Committee david_sinclair@hms.harvard.edu
Professor of Genetics, Harvard Medical School

Dr. Matt Kaeberlein, Advisor kaeber@uw.edu
Professor of Pathology, University of Washington